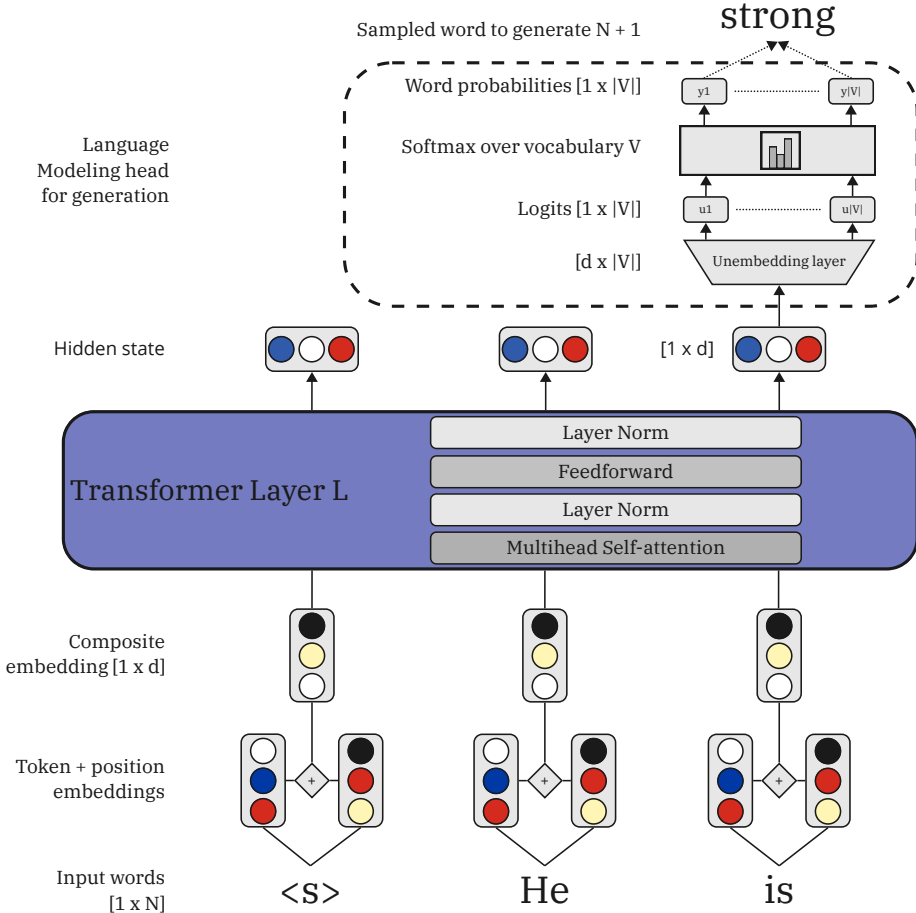




$$P(\text{strong} \mid \text{He is}) = \arg \max_i y_i$$

$$y = \text{softmax}(u)$$

$$u = h_N^L E^T$$

h_N^L – last hidden state of L-th transformer block

E^T – undembedding matrix

Transformers work in the pre-train and fine-tune paradigm. The pre-trained transformer can solve different tasks with just changing its head (transfer learning).

Examples (encoder): BERT (2018)

Examples (decoder): GPT-2 (2019)

Examples (encoder-decoder): T5 (2020)